

AegisVault-J

A Secure Encrypted Container for Sensitive Files

Group – 6

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Problem

In the digital era, users store highly sensitive information such as:

- personal and academic documents
- financial records and credentials
- private files and encryption keys

However, users still experience:

- accidental data exposure
- weak folder-based protection
- excessive reliance on OS security

Core problem: data becomes vulnerable once the system is unlocked.

Gap in Existing Systems

Most existing security solutions:

- encrypt entire disks instead of selected data
- require OS-level mounting or privileges
- expose files immediately after authentication
- rely heavily on user awareness

They fail to ensure:

- strict data movement boundaries
- controlled internal file access

Result: security weakens post-login.

Solution: AegisVault-J

AegisVault-J is a secure encrypted container that:

- stores all sensitive data inside a single vault
- uses modern cryptographic standards
- never stores user passwords
- enforces explicit import and export actions

Objective: protect data even after authentication.

How AegisVault-J Works

AegisVault-J follows a controlled security pipeline:

- ① Vault creation by the user
- ② Secure key derivation using Argon2id
- ③ Explicit file import into vault
- ④ Encryption using AES-256-GCM
- ⑤ Controlled file export on demand

Privacy guarantee: no plaintext or passwords are stored.

Virtual File System

AegisVault-J implements a logical virtual file system:

- folder-based structure inside the vault
- independent of OS file system
- no disk-level encryption required
- no OS-level mounting

Advantage: structured security without system exposure.

Benefits

AegisVault-J offers:

- prevention of accidental data leakage
- strong cryptographic security
- cross-platform portability
- user-controlled data flow

Value delivered: complete ownership of sensitive data.

Future Scope

Possible future enhancements include:

- hardware-backed key protection
- encrypted cloud vault synchronization
- detailed access audit logs
- enterprise-level deployment support

Conclusion

AegisVault-J redefines secure storage by making it:

- explicit
- controlled
- portable

It shifts security from the OS level to the application level using modern cryptography.

Thank You

Questions & Discussion